CALIFORNIA
Water Boards

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Secretary for
Environmental Protection

Central Valley Regional Water Quality Control Board

9 March 2018

Art de Werk
City Manager
City of Atwater
755 Bellevue Road
Atwater, CA 95301

NO FURTHER ACTION DETERMINATION, CITY OF ATWATER'S FORMER SLUDGE DRYING BEDS (PARCEL 056-241-007-000), FORMER WASTEWATER TREATMENT FACILITY, 550 COMMERCE DRIVE, ATWATER, MERCED COUNTY

The City of Atwater owns property at 550 Commerce Avenue in Atwater that formerly contained the City of Atwater's wastewater treatment facility. The 550 Commerce property is comprised of three parcels (Parcel numbers 003-180-012-000, 056-241-005-000, and 056-241-007-000). The City of Atwater plans to retain ownership of parcel 003-180-012-000 (northwest corner) and the southern portion of parcel 056-241-005-000 that contains a wastewater pump station that is still in use. The City of Atwater plans to sell the northern portion of parcel number 056-241-005-000 and is in discussions with Jim Brisco Enterprises, Inc. on selling or leasing parcel number 056-241-007-000. The site is proposed to be developed as a batch plant.

On 19 June 2017, Central Valley Regional Water Quality Control Board (Central Valley Water Board) staff received a technical report *Former Wastewater Treatment Facility Biosolids Drying Beds, Merced County, 2016 - 2017 Groundwater Risk Assessment and Mitigation Report (*Risk Assessment) prepared by West Yost Associates (Yost) on behalf of the City of Atwater. The Risk Assessment was signed and certified with the stamp of Mr. James F. Strandberg, a California Certified Engineering Geologist (CEG 1685).

The Risk Assessment Report characterizes the soil and groundwater conditions that currently exist under the former sludge drying beds and wastewater treatment facility. The Risk Assessment Report concludes that the results from mitigation and monitoring activities conducted at the site (excavation and removal of surficial soils and groundwater monitoring) indicate that the remaining nitrogen in the soils underlying the former sludge drying beds will not pose a threat to the quality of the underlying groundwater, provided the proposed mitigation measures are enacted. Nitrate as nitrogen is present in groundwater at concentrations exceeding the Primary maximum contaminant level of 10 milligrams per liter, but the results are stable and do not indicate increasing concentrations. The site will be capped with concrete to minimize the infiltration of water from the surface through the vadose zone to the underlying groundwater. The City of Atwater will develop the site in phases over the next three to five years, and will submit annual site maintenance and inspection reports. The City of Atwater will continue to conduct groundwater monitoring to ensure the nitrate as nitrogen concentrations are remaining stable and do not pose a threat to any nearby domestic or irrigation wells.

Central Valley Water Board staff has reviewed the Risk Assessment and associated reports, and concur that the mitigation activities at the site, proposed post-closure development plans, and groundwater monitoring have reduced or will reduce the threat to the underlying groundwater quality.

Central Valley Water Board staff do not object to the City of Atwater proceeding with its proposed site redevelopment of parcel 056-241-007-000 as described in the Risk Assessment Report. We intend to issue a Monitoring and Reporting Program under separate cover that will require regular site inspections, groundwater monitoring, and the submittal of an annual report.

If you have any questions regarding this matter, please contact Jeff Pyle at (559) 445-5145 or via email at jpyle@waterboards.ca.gov.

Clay L. Rodgers

for Pamela C. Creedon
Executive Officer

cc: Kathryn E. Gies, West Yost Associates, Walnut Creek, CA

Enclosure: Technical Memorandum, No Further Action Determination, City of Atwater's Former Sludge Drying Beds (Parcel 056-241-007-000), Former Wastewater Treatment Facility, 550 Commerce Drive, Atwater, Merced County.

TECHNICAL MEMORANDUM

TO: Matt. S. Scroggins
Senior Water Resource Control Engineer

Scott J. Hatton
Supervising Water Resource Control Engineer

Clay L. Rodgers
Assistant Executive Officer

FROM: Jeffrey S. Pyle
Engineering Geologist
PG No. 7375

DATE: 9 March 2018

NO FURTHER ACTION DETERMINATION, CITY OF ATWATER'S FORMER SLUDGE DRYING BEDS (PARCEL 056-241-007-000), FORMER WASTEWATER TREATMENT FACILITY, 550 COMMERCE DRIVE, ATWATER, MERCED COUNTY

The City of Atwater's former wastewater treatment facility (WWTF) property is at 550 Commerce Drive in the south-central portion of Atwater as shown in Attachment A. The discharge was regulated by Waste Discharge Requirements (WDRs) R5-2007-0063 (NPDES CA0079197), which was rescinded in October 2012, after a new regional wastewater treatment plant became operational. Wastewater treatment occurred at the site for years (the earliest WDRs are listed as 73-087), and there are/were soil and groundwater impacts from the operations of the WWTF.

BACKGROUND

WDRs R5-2007-0063 allowed for a discharge of up to 6.0 million gallons per day (mgd) of disinfected secondary treated wastewater into the Atwater Drain, a water of the United States. Prior to the closing of the former WWTF in June 2012, the WWTF was producing an average of about 3.5 mgd of disinfected secondary treated wastewater and about 700 tons of Class B biosolids annually. The sludge was discharged to 10-unlined sludge drying beds.

Closure activities have included the submittal of several technical reports prepared by West Yost Associates (Yost) on behalf of the City of Atwater to evaluate the quality of the underlying soil and groundwater. A list of the pertinent technical reports and actions relative to closure of the former Atwater WWTF are summarized in the following table.

Table 1

Closure Activities for Former Atwater WWTF	
Date	**Report/Action**
May 2012	City submits Original Closure Plan.
June 2012	Discharge ceased to former WWTF.
May 2013	City submits revised Closure Plan.
June 2013	Central Valley Water Board staff issue letter of conditional approval.
September 2013	City removes biosolids from drying beds.
December 2013	City submits Progress Report.
March 2014	City conducts additional soil sampling.
April 2014	City submits additional March soil sampling results and collects more samples in April.
May 2014	City submits additional April soil sampling results.
December 2014	City submits Nitrate Risk Assessment and Mitigation Report.
March 2015	City submits a revised Nitrate Risk Assessment and Mitigation Report.
June 2015	Central Valley Water Board staff provided written acceptance of the revised Nitrate Risk Assessment and Mitigation Report and requests implementation of proposed soil excavation and submittal of a groundwater monitoring well installation plan.
August 2015	City submits soil excavation and soil confirmation results.
August 2015	City submits a Groundwater Monitoring Well Installation Work Plan.
October 2015	Central Valley Water Board staff provide written acceptance of Groundwater Monitoring Well Installation Work Plan
April 2016	City submits Monitoring Well Installation Report with 1st Quarter 2016 Groundwater Monitoring Results.
December 2016	City submits 2016 Annual Groundwater Monitoring Report.
June 2017	City submits Technical Report for Clean Closure of former WWTF.
June 2017	City submits 2016-2017 Groundwater Risk Assessment Report.
July 2017	City submits 2nd Quarter 2017 Groundwater Monitoring Report.
September 2017	City submits 3rd Quarter 2017 Groundwater Monitoring Report.

WWTF Property

The WWTF property is comprised of three parcels as follows:

- Parcel 003-180-012-000,
- Parcel 056-241-005-000,
- Parcel 056-241-007-000.

I have included Attachment B that shows the three parcels and the location of the former WWTF, effluent storage ponds, and the sludge drying beds. The City plans to retain ownership of parcel 003-180-012-000 (northwest corner) and the southern portion of parcel 056-241-005-000 as shown on Attachment B, but plans to sell the northern portion of parcel 056-241-005-000 and are in discussions on selling or leasing parcel 056-241-007-000 for commercial development (concrete batch plant).

Parcel 056-241-005-000 contained the treatment facilities that consisted of a headworks, two primary clarifiers, three aeration basins, four secondary clarifiers, three chlorine contact chambers, and three unlined emergency storage ponds. Parcel 056-241-007-000 contained the ten sludge drying beds. Parcel 003-180-012-000 did not include facilities related to the WWTF.

Parcel 056-241-005-000 – WWTF and Emergency Storage Ponds
The Discharger collected soil samples from the emergency storage ponds in 2013. Soils underlying the storage ponds had ammonia and total Kjeldahl nitrogen (TKN) results that were about two to three times the values observed in the background sample, but metal concentrations were similar to background. While the nitrogen and ammonia values in the soil samples collected from the storage ponds exceed the background values, the values were low and the 2017 Closure Summary notes the parcel will be graded (former ponds filled and compacted) to minimize the percolation of rainwater into the underlying soils. The reported concentrations were compared to Title 27 total designated levels (TDLs) and soluble designated levels (SDLs), and the comparison found that all concentrations are less than the Title 27 TDLs and SDLs. The Discharger will collect and analyze some soil samples in the areas of the decommissioned WWTF, and submitted a Work Plan to collect soil samples on 31 January 2018. Final closure of Parcel 056-241-005-000 will be addressed separately from the closure of the sludge drying beds.

CLOSURE OF SLUDGE DRYING BEDS
Parcel 056-241-007-000 – Sludge Drying Beds
I reviewed several technical reports and data contained in the file record (Table 1), but the following technical reports prepared by Yost on behalf of the City of Atwater provided most of the data for my review and recommendations.

- *Closure of City of Atwater Former Wastewater Treatment Facility Biosolids Drying Beds, Revised Nitrate Risk Assessment and Mitigation Report, March 2015 (Revised Report).*
- *Former Wastewater Treatment Facility Biosolids Drying Beds, Merced County, 2016 - 2017 Groundwater Risk Assessment and Mitigation Report. 17 June 2017* (2017 Risk Assessment).

Soil Impacts
In efforts to close Parcel 056-241-007-000, the City removed biosolids and collected soil samples in September 2013. Discussions with Central Valley Water Board staff led to additional soil sampling in March and April 2014 to characterize the soils underlying the former sludge drying beds. The results of the soil sampling indicated soils with elevated total Kjeldahl nitrogen and nitrate as nitrogen concentrations were present in drying beds (DB) 1, 6, 7 and 8. Based on the results of the soil sampling, and on discussions with Central Valley Water Board staff, the

City agreed to remove a foot of soil from DB 1, DB 6, and DB 8, and two feet of soil from DB 7. Table 2 summarizes the averages of soil sampling results from the 2013 and 2014 sampling events and compares those results to the 2015 confirmation soil-sampling event. Samples AS-2 and AS-3 were collected as background soil samples from outside of the sludge drying beds east/southeast of DB-10 (Attachment B).

Table 2
Drying Bed Soil Sampling Results for Total Kjeldahl Nitrogen[1]

Date	Drying Bed No.1		Drying Bed No. 6		Drying Bed No. 7			Drying Bed No. 8		Background AS-2			Background AS-3		
	Depth – feet bgs[2]		Depth – feet bgs[2]		Depth – feet bgs[2]			Depth – feet bgs[2]		Depth – feet bgs[2]			Depth – feet bgs[2]		
	0-1	1-2	0-1	1-2	0-1	1-2	2-3	0-1	1-2	0-1	1-2	2-3	0-1	1-2	2-3
Sep 13	960	---	610	---	2200	---	---	1800	---	500	140	98	390	150	110
Mar 14	---	240	---	480	---	2600	---	---	460	---	---	---	---	---	---
Apr 14	---	---	---	320	---	2900	---	---	---	---	---	---	---	---	---
Jul 15	---	**620**	---	**480**	---	---	**480**	---	**630**	---	---	---	---	---	---

1. All results are in milligrams per kilogram
2. bgs = below the ground surface.

In Table 2 above, the shaded areas indicate areas where soil was excavated and removed and the values shown within the shaded areas represent the average TKN concentrations of several soil samples (two to four) collected in 2013 to characterize the soil quality in that drying bed. The values shown in bold text are the average TKN concentration of the confirmation soil sampling conducted in 2015 after the excavation of the overlying soils. While the results show that, some soil remains with TKN values that exceed background values, the confirmation soil sampling results are less than the previous results from the areas where soil was excavated.

Parcel 056-241-007-000 – Groundwater Monitoring

Following the confirmation soil sampling, Central Valley Water Board staff requested that groundwater monitoring wells be installed to characterize any potential groundwater degradation that may have occurred from the operation of the former WWTF. The City installed three groundwater monitoring wells in March 2016 and have generally conducted quarterly monitoring of the wells since that time. Initially, groundwater sampling was conducted for one year or four quarters, but the Discharger has continued monitoring the groundwater at the request of Central Valley Water Board staff.

The depth to water has averaged from about 80 feet bgs to 90 feet bgs with a flow direction that is generally to the west/northwest, but the direction of groundwater flow is variable and influenced by nearby irrigation and supply wells. The direction of groundwater flow was to the southeast in the Second Quarter of 2017, and the likely cause was a City of Atwater supply well located about 1,200 feet southeast of the former WWTF. Several irrigation wells are northwest of the former WWTF and they likely influence the direction of groundwater flow when they are on.

The west/northwest flow direction appears to indicate wells MW-2 and MW-3 are upgradient wells and MW-1 is downgradient, and this is generally correct. However, the frequently changing direction of groundwater flow as noted in the quarterly groundwater monitoring events has resulted in the nitrogen from the sludge drying beds causing degradation of the underlying groundwater in all three of the wells. The nitrate as nitrogen values in groundwater are discussed in detail following the introduction of the groundwater data in Table 3.

The City has submitted seven groundwater monitoring reports starting with March 2016 (First Quarter 2016) and extending through December 2017 (Fourth Quarter 2017), and the results and corresponding averages are shown in Table 3. The values shown in bold for electrical conductivity and total dissolved solids exceed the recommended Secondary maximum contaminant levels (MCL) for each constituent and the values shown in bold for nitrate as nitrogen exceed the Primary MCL for nitrate as nitrogen.

Table 3
Groundwater Sampling Results

Well	Date	Electrical Conductivity umhos/cm[1]	Total Dissolved Solids mg/L[2]	Nitrate as Nitrogen mg/L[2]	Total Kjeldahl Nitrogen mg/L[2]	Total N Nitrogen mg/L[2]
MW-1	3/7/16	546	380	8.4	0	8.4
	7/28/16	546	410	9.3	0.3	9.6
	10/18/16	640	---	---	---	---
	1/5/17	437	350	8	0	8
	5/11/17	479	370	7.4	0.05	7.5
	7/27/17	---	---	8.7	---	---
	12/12/17	432	350	7.7	0	7.7
Averages		513	372	8.2	0.1	8.2
MW-2	3/7/16	780	**510**	10	0	10
	7/28/16	509	410	**17**	0.26	17.3
	10/18/16	495	380	**17**	0.12	17.1
	1/5/17	465	390	**16**	0	16
	5/11/17	542	420	**14**	0.04	14
	7/27/17	545	420	**16**	0	16
	12/12/17	483	390	**17**	0.12	17.1
Averages		546	417	**15**	0.08	15
MW-3	3/7/16	**923**	**530**	**15**	0.05	15.1
	7/28/16	583	395	**13**	0.13	13.1
	10/18/16	492	375	**14**	0.03	14
	1/5/17	480	390	**18**	0.03	18
	5/11/17	495	410	**17**	0.1	17.1
	7/27/17	505	420	**15**	0.5	15.5
	12/12/18	446	370	**16**	0.16	16.2
Averages		580	420	**15**	0.14	15.5

1. umhos/cm = micromhos per centimeter.
2. mg/L = miiligrams per liter

The quarterly monitoring results indicate the water is of good quality with the exception of nitrate as nitrogen, which exceeds the Primary MCL of 10 mg/L in MW-2 and MW-3, and averages 8.2 mg/L in MW-1. All of the values from MW-2 and MW-3 have been at or above the MCL of 10 mg/L, but the trends in concentrations are stable and do not indicate an increasing trend of nitrate as nitrogen in groundwater. The 2017 Risk Assessment indicates that the regional nitrate as nitrogen results are similar to those in the site monitoring wells noting values of 6 to 12 mg/L at nearby Teasdale Foods facility, and concludes that the former sludge drying beds are not a significant source of nitrogen in groundwater.

I do concur that with the mitigation measures conducted and proposed by the City of Atwater (removal of degraded soils and biosolids from the former sludge drying beds, proposed capping of the site, and continued groundwater monitoring) there is or will be little if any threat to the underlying groundwater. However, I do not concur that the nitrate as nitrogen results in MW-2 and MW-3 are similar to background. The WWTF has been in operation for at least 45 years and as the current groundwater monitoring has shown, the direction of groundwater flow is highly variable. The Department of Water Resources Lines of Equal Elevation Maps for the Spring of 2008 and 2009 show the direction of flow as being to the northwest. The Teasdale Foods site is only about 250 feet north/northwest (directly across Highway 99) of the former WWTF. Considering how close the Teasdale site is to the WWTF and the fluctuating groundwater flow directions, it is not impossible for the WWTF to have contributed to the values reported at the Teasdale site. Furthermore, the March 2015 Risk Assessment by Yost includes Figure 8 that shows the known nitrate as nitrate concentrations in Domestic Wells within one mile of the WWTF. Seven wells are shown to the east and potentially are upgradient of the former WWTF and former sludge drying beds. Nitrate **as nitrate** concentrations were found to range from 19 mg/L to 35 mg/L in all seven of these wells. All of the values are less than the 45 mg/L MCL for nitrate as nitrate. The results correlate to **nitrate as nitrogen** results that range from 4.2 mg/L to 7.8 mg/L that are all less than the MCL of 10 mg/L, and well below the 15 mg/L average nitrate as nitrogen results observed in wells MW-2 and MW-3.

Proposed Site Redevelopment

The City of Atwater is in the process of finalizing an agreement with Jim Brisco Enterprises Inc., to redevelop parcel 056-241-007-000. A Site Redevelopment Plan was submitted as Appendix D of the 2015 Redevelopment Plan and included the following mitigation measures to limit or eliminate the potential migration of the residual nitrogen in soil beneath the former sludge drying beds.

- ***Surface Pavement*** *- The Site will be fully covered by buildings or pavement over the course of a three- to five-year buildout. The pavement will be four-inch thick asphalt concrete or Portland cement concrete over four-inch Class II aggregate base. Areas with truck traffic will have thicker pavement.*
- ***Water Use and Leak Detection*** *– Water use will occur in four primary areas or buildings including the concrete batch plant, crusher, office/showroom, and shop building. During Site redevelopment and prior to backfilling the water pipeline trenches, the water distribution system will be pressure tested to ensure the system is tight. The water distribution system will be pressure tested every five years to confirm no significant leakage.*

- ***Process Water Containment** – Water from the concrete plant will be processed in the 'reclaimer settling ponds' which are concrete-lined (i.e., floor and walls) flow-through cells to settle suspended solids. The water will be re-used at the concrete plant. A concrete-lined wash rack will contain truck wash water. Overflow from the wash rack will flow to a concrete-lined 'retention basin' on the southern boundary of the Site.*
- ***Storm Water Containment** – Storm water runoff for the entire Site will be directed to the onsite concrete-line 'retention basin' for containment. Flows exceeding the capacity of the 'retention basin' will be either directed to the City's sewer system or to the existing storm water retention basin located southwest of the Site (Figure 2). This proposed containment strategy is expected to be adequate for gaining exemption from the State's Industrial Storm water Permitting requirements. Specifically, the recently adopted NPDES General Permit for Storm Water Discharges Associated with Industrial Activities Order 2014-0057-DWQ NPDES No. CAS000001 (General Permit) includes 1) "no discharge" eligibility requirements that allow for exemption if the facility is engineered and constructed to contain storm water so that there will be no discharge of industrial storm water to waters of the United States and 2) a No Discharge Technical Report is filed with the State Water Board that demonstrates how the facility meets the eligibility requirements and is signed by a California licensed Professional Engineer.*
- ***Landscape Irrigation** – The trees surrounding the perimeter of the concrete plant will be watered with a drip irrigation system operated to provide the minimum amount of water needed to sustain the trees. Pressure gauges are proposed to detect any significant leaks from the drip irrigation system.*
- ***Excavation Practices** – A practice will be established to ensure that any section of pavement removed as part of an excavation will be replaced to match the original pavement type and thickness.*

Annual Site Inspection, Maintenance, and Groundwater Monitoring
To help ensure that the redevelopment plan strategies will continue to limit the infiltration of surface water into soils beneath the former WWTF over the long term, the Discharger proposes to perform regular inspection and maintenance procedures, and will submit a Site Maintenance Report annually. The Discharger has proposed that prior to the onset of the rainy season and no later than October 1, of each year; they will inspect the facility pavement for cracks that could potentially allow surface water to infiltrate into the soils beneath the pavement. The Discharger will use an epoxy joint filler to seal cracks identified during these inspections and any other similar cracks observed during the year to maintain the integrity of the pavement.

The Discharger originally agreed to conduct one year of groundwater monitoring, and subsequently agreed to continue groundwater monitoring for a second year. In discussions with Central Valley Water Board staff, the Discharger has agreed to continue groundwater monitoring of the three existing groundwater monitoring wells.

COMMENTS
The City has evaluated the soil and groundwater quality beneath the former Atwater WWTF and concluded that impacts to both the underlying soil and groundwater have occurred. However,

the City has or will implement mitigation measures to reduce and/or eliminate the potential for nitrates in the soil to further threaten the underlying groundwater quality.

The City has removed biosolids and soils with elevated nitrogen from the former sludge drying beds and will develop the site in phases to reduce the potential for water to transport nitrogen through the vadose zone to the underlying groundwater. Site development will include grading and capping the site, and constructing storm and process-water retention basins. Capping the site will minimize the potential for water to percolate through the vadose zone and degrade the underlying groundwater.

Groundwater monitoring results for MW-2 and MW-3 indicate nitrate as nitrogen concentrations exceed the Primary MCL of 10 mg/L in both wells. However, the concentrations are stable and do not indicate an increasing trend. Considering the following:

- The WWTF was taken offline in 2012 (eliminating any further addition of nitrogen to the soil);
- Soil with elevated nitrogen concentrations was excavated and removed from the sludge drying beds from 2013 through 2015; and
- The site will be developed, graded, capped, and maintained to minimize the potential for water to percolate through the underlying soils.

It is unlikely that the existing remaining soils would continue to contribute to the nitrate as nitrogen concentrations in the underlying groundwater that exceed the Primary MCL of 10 mg/L.

RECOMMENDATIONS

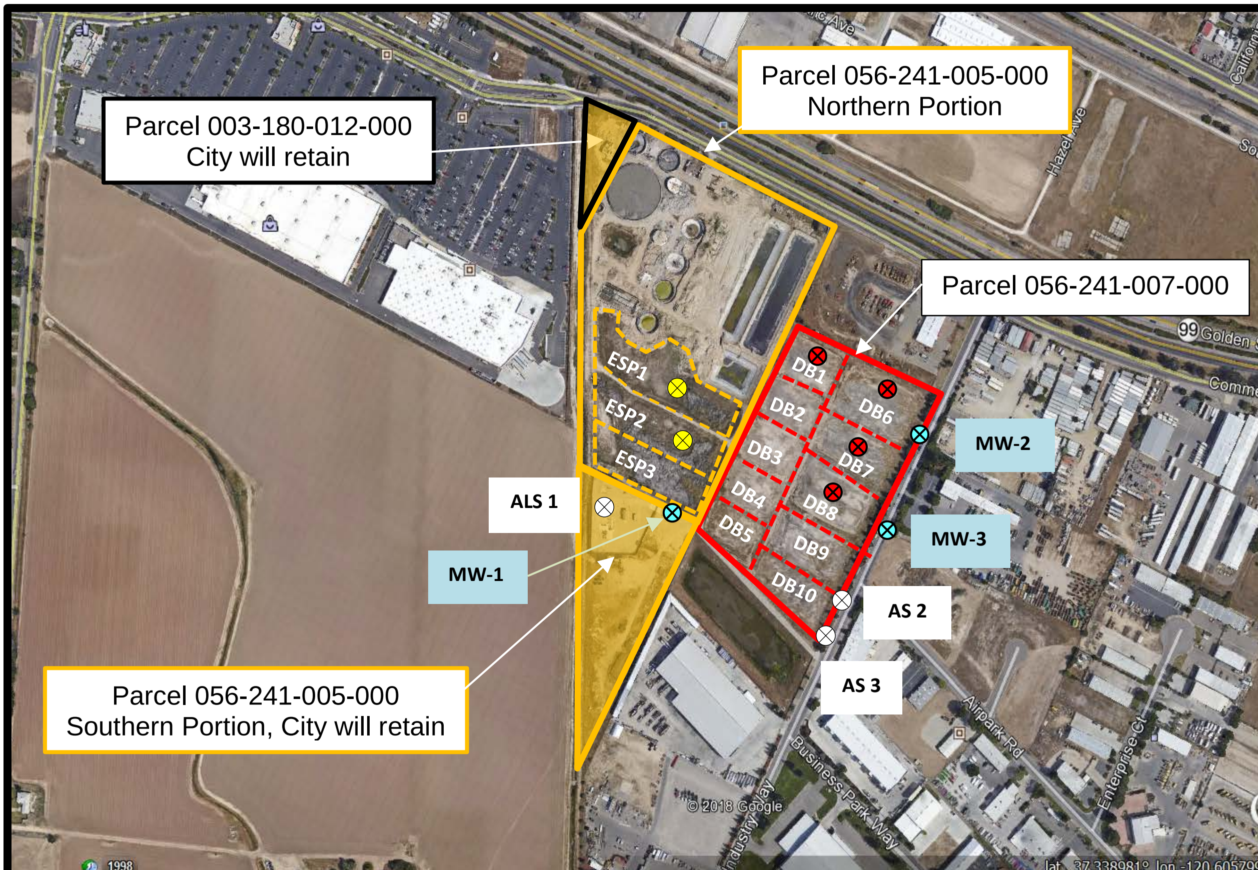
The closure request and the proposed development should be approved. To document that the Closure Plans are progressing as proposed, the City has proposed the submittal of an annual site inspection and maintenance report. The Annual reports would be due by 1 February of each year. Additionally, the Discharger will retain the three existing groundwater monitoring wells and continue groundwater monitoring to demonstrate that the nitrate as nitrogen concentrations in groundwater are remaining stable and are not a threat to any nearby supply or domestic wells.

SITE VICINITY MAP

NO FURTHER ACTION DETERMINATION
FORMER CITY OF ATWATER WWTF
FORMER WDR ORDER R5-2007-0063
MERCED COUNTY

Approximate Scale in Miles
0.0 0.25 0.50 0.75 1.0 1.25

ATTACHMENT A



LEGEND

- Parcel 003-180-012-000
- Parcel 056-241-005-000 (Northern Portion)
- Parcel 056-241-005-000 (Southern Portion)
- Parcel 056-241-007-000
- Effluent Pond Soil Sample
- Background Soil Sample
- Drying Bed Confirmation Soil Samples
- Groundwater Monitoring Wells

SITE MAP

NO FURTHER ACTION DETERMINATION
FORMER CITY OF ATWATER WWTF
FORMER WDR ORDER R5-2007-0063
MERCED COUNTY

0 250 500 750 1,000

Approximate Scale in Feet

Attachment B